

ITEP 414 – System Administration and Maintenance
Week 2

ENTERPRISE INFRASTRUCTURE PLAN

NexaCore Solutions

Prepared by	Gimang, Ary Y.
Program	Bachelor of Science in Information Technology
Course	ITEP 414 - System Administration and Maintenance
Project Type	Individual Portfolio Project
Company Size	20 Employees
Office	Santa Cruz, Laguna, Philippines

PART I - COMPANY PROFILE

1.1 Company Name

NexaCore Solutions

1.2 Nature of Business

NexaCore Solutions is a fictional software development company that provides customized software applications and digital technology solutions to businesses. The company focuses on developing reliable and user-friendly software while providing technical support and IT-related services to its clients.

1.3 Company Vision

To become a trusted technology partner that provides reliable, innovative, and accessible digital solutions to businesses.

1.4 Company Mission

NexaCore Solutions aims to provide quality software and IT services through skilled employees, modern technology, secure infrastructure, and excellent customer service.

1.5 Office Location

2nd Floor, Laguna Technology Center, Santa Rosa, Laguna, Philippines

(The office location is fictional and is used for this academic project.)

1.6 Organizational Structure

NexaCore Solutions has four main departments: Information Technology, Human Resources, Finance, and Sales. The company has a total of 20 employees.

Organizational Structure:

General Manager

Information Technology Department – 5 Employees

Human Resources Department – 4 Employees

Finance Department – 5 Employees

Sales Department – 6 Employees

1.7 Employee Distribution

Information Technology	5
Human Resources	4
Finance	5
Sales	6
Total	20

Company Profile Summary

NexaCore Solutions is designed as a small software development startup operating from a single office floor. Since the company is newly established, it currently requires a complete IT infrastructure, including computers, servers, networking equipment, software, security measures, and backup systems. The infrastructure plan will be designed according to the needs of its 20 employees and four departments.

PART II - HARDWARE INVENTORY

2.1 Hardware Inventory

NexaCore Solutions requires a variety of hardware to support the daily operations of its 20 employees. The selected equipment is based on the company's four departments and its need for reliable computing, networking, storage, printing, and backup capabilities.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	15	IT, HR, Finance, Sales	Provides employees with reliable workstations for daily business activities.
HW-002	Laptop	5	IT, Sales	Provides portable computers for IT work, meetings, presentations, and flexible work.
HW-003	Server	1	IT	Provides centralized services, file storage, and other

				internal IT functions.
HW-004	Router	1	IT	Connects the company's internal network to the internet and manages network traffic.
HW-005	Managed Switch	1	IT	Connects computers, servers, printers, and other wired devices to the network.
HW-006	Network Printer	2	HR, Finance	Allows employees to print documents, reports, forms, and other business materials.
HW-007	UPS	5	IT	Provides temporary power and protects important equipment from sudden power interruptions.
HW-008	Wireless Access Point	3	All Departments	Provides wireless network connectivity throughout the office.
HW-009	NAS Storage	1	IT	Provides centralized network storage and supports data backup.
HW-010	External Backup Drive	2	IT	Provides additional offline copies of important company data.
HW-011	Monitor	20	All Departments	Provides displays for employee workstations.

2.2 Hardware Selection Justification

The hardware requirements are based on the company's 20 employees and four departments. Fifteen desktop computers and five laptops provide 20 primary workstations for employees. Laptops are assigned where portability is more useful, particularly for IT personnel and Sales employees.

One server is included to provide centralized IT services, while the router and managed switch form the foundation of the company's wired network. Two network printers are provided for document-related tasks, particularly for Human Resources and Finance.

Three wireless access points are recommended to provide wireless connectivity across the single office floor. A NAS device and two external backup drives provide additional storage and backup options. UPS units are included to help protect important IT equipment from power interruptions.

Overall, the selected hardware provides NexaCore Solutions with the basic infrastructure required to operate its business while allowing room for future expansion.

PART III – SOFTWARE INVENTORY

3.1 Software Inventory

NexaCore Solutions requires software that supports office productivity, software development, system administration, security, communication, and file management. The selected software is based on the company's business operations and the responsibilities of its different departments.

Software	Version	License	Purpose
Windows 11 Pro	Current Supported Version	Commercial	Provides the primary operating system for employee workstations.
Ubuntu Server	24.04 LTS	Open Source	Provides the operating system for the company's server.
Microsoft Office	Microsoft 365	Subscription	Supports document creation, spreadsheets, presentations, and other office tasks.
Visual Studio Code	Current Stable Version	Free	Provides a code editor for software development and programming tasks.
Git	Current Stable Version	Open Source	Provides version control for software development projects.
VirtualBox	Current Stable Version	Free/Open Source	Allows IT personnel

			to create and manage virtual machines for testing.
Google Chrome	Current Stable Version	Free	Provides web browsing and supports web application testing.
Microsoft Defender	Included with Windows	Included	Provides security and protection against malware and other threats.
AnyDesk	Current Version	Commercial/Free Options	Allows authorized IT personnel to provide remote technical support.
7-Zip	Current Stable Version	Open Source	Provides file compression and extraction capabilities.
GitHub Desktop	Current Version	Free	Provides a graphical interface for managing Git and GitHub repositories.

3.2 Software Requirements and Justification

Windows 11 Pro

Windows 11 Pro will be used as the main operating system for employee workstations. It provides the environment needed for office applications, communication, web browsing, and other business activities.

Ubuntu Server

Ubuntu Server will be installed on the company's server. It provides a Linux-based server environment that can support centralized services and internal infrastructure.

Microsoft Office

Microsoft Office will be used by employees for creating documents, preparing spreadsheets, developing presentations, and performing other office-related tasks.

Visual Studio Code

Visual Studio Code will be primarily used by the Information Technology department for programming, software development, and editing source code.

Git

Git will be used by the IT department to track changes in source code and manage software development projects.

GitHub Desktop

GitHub Desktop provides an easier graphical interface for employees who need to work with Git and GitHub repositories.

VirtualBox

VirtualBox allows IT personnel to create virtual machines. These virtual environments can be used for testing operating systems, applications, and configurations without affecting the main systems.

Google Chrome

Google Chrome will be used for internet access, research, web-based business applications, and testing web applications developed by the IT department.

Microsoft Defender

Microsoft Defender will provide basic security protection for Windows workstations. It helps protect company computers from malware and other security threats.

AnyDesk

AnyDesk can be used by authorized IT personnel to provide remote technical assistance when troubleshooting employee computers. Access should be restricted to authorized personnel.

7-Zip

7-Zip will be used to compress and extract files. This can help employees manage large files and organize project files more efficiently.

3.3 Software Summary

The selected software provides NexaCore Solutions with the tools needed for daily business operations, software development, system administration, security, and file management. The combination of commercial, subscription-based, free, and open-source software allows the company to meet its requirements while considering cost and functionality.

PART IV – NETWORK INVENTORY

4.1 Network Equipment Inventory

NexaCore Solutions requires reliable networking equipment to connect its 20 employees, server, printers, and other network devices. The network infrastructure will provide internet connectivity, wired communication, wireless access, and basic network security.

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Connects the company's network to the internet service provider.
NET-002	Business Router	1	Routes network traffic between the

			company's internal network and the internet.
NET-003	Firewall	1	Helps protect the company's internal network from unauthorized network traffic and external threats.
NET-004	Managed Switch	1	Connects wired computers, the server, printers, and other network devices.
NET-005	Wireless Access Point	3	Provides wireless network connectivity throughout the office.
NET-006	Patch Panel	1	Organizes and manages Ethernet cable connections in the network area.
NET-007	CAT6 Ethernet Cable	1 Box	Provides reliable wired connections between network devices and workstations.
NET-008	RJ45 Connectors	50 Pieces	Used to terminate CAT6 Ethernet cables for wired network connections.

4.2 Network Equipment Justification

ISP Modem

One ISP modem is required to establish the company's connection to its internet service provider. It serves as the connection point between the external internet service and the company's network infrastructure.

Router

One business router will manage traffic between the internal company network and the internet. It will serve as an important component of the company's network infrastructure.

Firewall

A dedicated firewall will be used to help protect the internal network from unauthorized traffic and external threats. It provides an additional layer of security between the company's network and the internet.

Managed Switch

A managed switch will connect the company's wired devices, including employee computers, the server, printers, and other network equipment. A managed switch also provides better control and management of the internal network.

Wireless Access Points

Three wireless access points are recommended to provide wireless coverage throughout the single office floor. Multiple access points can help provide more consistent wireless connectivity for employees and mobile devices.

Patch Panel

One patch panel will be used to organize the Ethernet cables entering the network area. This makes cable management easier and helps simplify future troubleshooting and maintenance.

CAT6 Ethernet Cables

One box of CAT6 Ethernet cable will provide the physical connections required for the company's wired network. CAT6 cabling is suitable for connecting computers, servers, printers, switches, and other network devices.

RJ45 Connectors

Fifty RJ45 connectors will be prepared for terminating Ethernet cables. Additional connectors are included to account for spare connections, replacement cables, and possible future network expansion.

4.3 Network Infrastructure Summary

The network infrastructure is designed to provide NexaCore Solutions with reliable wired and wireless connectivity. Internet traffic will enter through the ISP modem and pass through the router and firewall before reaching the internal network switch. The switch will then connect the server, employee workstations, printers, and wireless access points. This design provides a structured foundation for the company's current operations while allowing the network to be expanded as the company grows.

PART V – NETWORK DIAGRAM

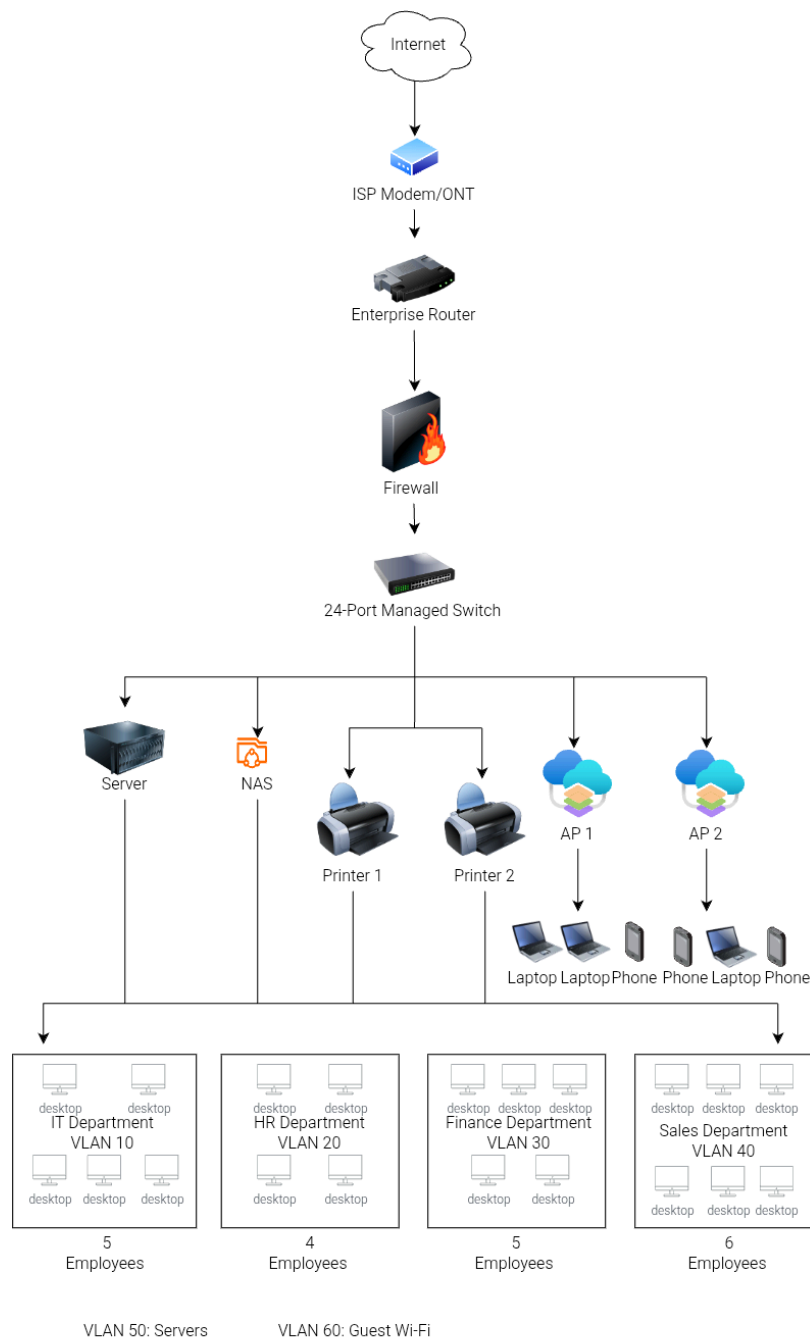
5.1 Network Topology

The proposed network topology for NexaCore Solutions is designed to provide reliable and organized communication between the company's internet connection, network infrastructure, servers, workstations, printers, and wireless devices.

The network follows a structured design in which the internet connection passes through the ISP modem, router, and firewall before reaching the managed switch. The managed switch then provides connections to the server, departmental workstations, printer, and wireless access point.

5.2 Enterprise Network Diagram

NexaCore Solutions – Enterprise Network Infrastructure



Prepared by: Gimang, Ary Y.
ITEP 414 – System Administration and Maintenance

5.3 Network Design Explanation

The network begins with the internet connection provided by the Internet Service Provider. The ISP modem receives the internet service and passes the connection to the router. The router manages traffic between the external network and the company's internal network.

The firewall is positioned between the router and the internal network to provide an additional layer of security. After passing through the firewall, network traffic reaches the managed switch. The switch connects the company's server, departments, printer, and wireless access point.

The four departments—Information Technology, Human Resources, Finance, and Sales—are connected to the internal network. The wireless access point provides wireless connectivity for authorized devices, while the server provides centralized IT services and storage. This design provides NexaCore Solutions with an organized network infrastructure that can be maintained and expanded as the company grows.

PART VI – SYSTEM ADMINISTRATION ROLES

6.1 Helpdesk Technician

Responsibilities

A Helpdesk Technician provides first-level technical support to employees. Their responsibilities include troubleshooting hardware and software problems, assisting users with computer-related concerns, installing and configuring workstations, managing basic user issues, documenting technical incidents, and escalating complex problems to other IT personnel.

Skills

A Helpdesk Technician should have good communication and customer service skills, basic hardware and software troubleshooting abilities, knowledge of operating systems, basic networking knowledge, problem-solving skills, and patience when assisting users.

Common Tools

Common tools include Windows troubleshooting utilities, remote support applications, ticketing systems, Microsoft Defender, and system diagnostic tools.

Certifications

A suitable entry-level certification is CompTIA A+, which is commonly associated with foundational IT support and troubleshooting skills.

6.2 Network Administrator

Responsibilities

A Network Administrator is responsible for maintaining the company's network infrastructure. Their duties include configuring and maintaining routers and switches, monitoring network performance, troubleshooting connectivity problems, managing wireless networks, maintaining network security, and ensuring that employees can reliably access required network resources.

Skills

A Network Administrator should have knowledge of TCP/IP, IP addressing, routing and switching, network troubleshooting, network security, VLANs, and wireless networking. Good analytical and problem-solving skills are also important.

Common Tools

Common tools include Cisco IOS, network monitoring software, ping, traceroute, packet analysis tools, and Cisco Packet Tracer.

Certifications

The Cisco Certified Network Associate (CCNA) is an appropriate certification for this role. Cisco states that the CCNA covers network fundamentals, network access, IP connectivity, IP services, security fundamentals, automation, and programmability.

6.3 Linux System Administrator

Responsibilities

A Linux System Administrator manages and maintains Linux-based servers. Their responsibilities include installing and configuring Linux systems, managing users and permissions, maintaining server services, applying system updates, monitoring system performance, securing servers, performing backups, and troubleshooting server problems.

Skills

A Linux System Administrator should have knowledge of the Linux command line, file permissions, user management, networking, shell scripting, package management, server security, and system monitoring.

Common Tools

Common tools include Ubuntu Server, Bash, SSH, systemd, APT, and Linux system monitoring utilities.

Certifications

A relevant certification option is the ****Canonical Academy SysAdmin qualification****. Canonical's SysAdmin track includes Linux Terminal, Ubuntu Desktop, and Ubuntu Server administration, with the Ubuntu Server portion covering common server administration and security tasks.

6.4 Cloud Administrator

Responsibilities

A Cloud Administrator manages and monitors an organization's cloud-based resources and services. Their responsibilities include configuring cloud resources, managing cloud storage and computing services, monitoring cloud systems, managing access permissions, supporting cloud security, monitoring resource usage and costs, and assisting with backup and recovery.

Skills

A Cloud Administrator should understand cloud computing, networking, storage, virtualization, security, identity and access management, monitoring, and basic cloud cost management.

Common Tools

Common tools include the AWS Management Console, Microsoft Azure Portal, cloud command-line tools, virtual machines, cloud monitoring services, and cloud storage services.

Certifications

The AWS Certified Cloud Practitioner is an appropriate foundational certification. AWS describes it as validating foundational knowledge of AWS Cloud, services, and terminology. Its current exam covers cloud concepts, security and compliance, cloud technology and services, and billing, pricing, and support.

6.5 Collaboration Among the Four Professionals

The four IT professionals work together to keep NexaCore Solutions' technology infrastructure reliable, secure, and available to employees. The Helpdesk Technician is usually the first point of contact when an employee experiences a technical problem. If the issue involves network connectivity, it can be escalated to the Network Administrator. Problems involving Linux servers can be handled by the Linux System Administrator, while issues involving cloud-based services can be handled by the Cloud Administrator. The professionals communicate with one another when a problem affects multiple systems. For example, an employee may report that they cannot access a company application. The Helpdesk Technician can investigate the initial problem, the Network Administrator can check network connectivity, and the Linux or Cloud Administrator can investigate the application or server hosting the service. Working together allows the company to resolve technical problems efficiently while maintaining the availability and security of its IT infrastructure.

PART VII – INFRASTRUCTURE RECOMMENDATIONS

7.1 Internet Provider

NexaCore Solutions should use a reliable business-grade fiber internet connection that can support the activities of its 20 employees. A stable internet connection is important because the company depends on online communication, software development, cloud-based services, web applications, and file sharing.

The company should also consider having a secondary internet connection as a backup. This would help maintain business operations if the primary internet connection becomes unavailable.

7.2 Server Specifications

For the company's initial server, the following specifications are recommended:

Component	Recommended Specification
Processor	8-Core Server Processor
Memory	32 GB RAM
Storage	2 × 1 TB SSD
Storage Configuration	RAID 1
Network Gigabit Ethernet	Gigabit Ethernet
Operating System	Ubuntu Server 24.04 LTS
Power Protection	UPS
Backup	Automated backup to NAS and external storage

The server should provide sufficient resources for the company's initial operations. RAID 1 can provide redundancy by maintaining a mirrored copy of data across two drives. The server should also be connected to a UPS to help protect it from sudden power interruptions.

7.3 Backup Strategy

NexaCore Solutions should follow the 3-2-1 backup strategy to reduce the risk of losing important company data.

The strategy includes:

3 copies of important data

2 different types of storage

1 copy stored separately from the main system

The company's NAS can be used for centralized backups, while external backup drives can provide an additional offline copy. Important files should be backed up regularly, and backup restoration should be tested periodically to ensure that the data can be recovered when needed.

7.4 Security Recommendations

NexaCore Solutions should implement several security measures to protect its systems and company information. These include:

NexaCore Solutions should implement several security measures to protect its systems and company information. These include:

- Install and properly configure a firewall.
- Keep operating systems and applications updated.
- Restrict administrative access to authorized IT personnel.
- Use individual user accounts.
- Enable multi-factor authentication where available.
- Regularly back up important data.
- Update network equipment firmware.
- Monitor unusual network activity.
- Train employees to recognize phishing attempts and suspicious files.
- Apply appropriate access permissions to company files and systems.

Security policies should be reviewed regularly as the company grows and its infrastructure changes.

7.5 Antivirus and Endpoint Protection

Windows workstations should have Microsoft Defender enabled and regularly updated. IT personnel should monitor security alerts and ensure that endpoint protection remains active.

Employees should also be instructed not to disable security software or install unauthorized applications. Regular operating system and security updates should be applied to reduce exposure to known vulnerabilities.

7.6 Password Policy

NexaCore Solutions should establish a password policy that requires employees to:

- Use strong and unique passwords.
- Avoid sharing passwords with other employees.
- Never reuse company passwords for unrelated accounts.
- Enable multi-factor authentication when available.
- Immediately report compromised accounts or passwords.

- Avoid storing passwords in publicly accessible locations.
- Use an approved password manager if provided by the company.

Administrative accounts should have stronger access controls and should only be used by authorized IT personnel.

7.7 Infrastructure Expansion Plan

The infrastructure should be designed to support future growth. Although NexaCore Solutions currently has 20 employees, the company should prepare for an increase in its workforce and technology requirements.

Future expansion can include:

- Adding additional network switches.
- Increasing server storage and memory.
- Adding more wireless access points.
- Increasing NAS and backup capacity.
- Purchasing additional computers and laptops.
- Expanding cloud storage and computing resources.
- Implementing additional network security measures.
- Adding a secondary internet connection.
- Increasing network capacity as the number of employees grows.

The infrastructure should ideally be capable of supporting approximately 40 employees without requiring a complete redesign of the network.

7.8 Overall Recommendation

The proposed infrastructure provides NexaCore Solutions with a practical starting point for its operations. The combination of reliable hardware, appropriate software, structured networking, regular backups, security controls, and an expansion plan can help the company maintain efficient and secure IT operations.

The infrastructure should also be regularly reviewed by the IT team to identify outdated equipment, security risks, performance problems, and future technology requirements.

PART VIII – PERSONAL REFLECTION

Working on the Enterprise Infrastructure Planning project helped me understand that system administration involves more than simply fixing computers or configuring networks. I learned that proper planning is an important part of building an effective IT infrastructure. Before implementing any system, a System Administrator needs to understand the organization's requirements, employees, departments, equipment, software, and future goals.

One of the most important things I learned from this project was how hardware, software, and network components work together. Creating the hardware and software inventories helped me understand that every piece of equipment should have a specific purpose. I also learned that technology should be selected based on the organization's actual needs, available resources, security requirements, and potential future growth.

The most challenging part of the project for me was creating the enterprise network diagram. I had to determine how the internet connection, modem, router, firewall, switch, server, access points, printers, and different departments should be connected. I also had to make sure that the diagram was organized, clearly labeled, and consistent with the hardware and network inventories. Creating the diagram helped me better understand how different network devices work together within an organization.

This project also showed me why infrastructure planning is important before deployment. Without proper planning, an organization may purchase equipment that does not meet its needs, create an inefficient network, or fail to protect important information. Planning also makes it easier for an organization to expand because the infrastructure can be designed with future growth in mind.

Overall, this project improved my understanding of IT infrastructure planning, network design, hardware and software selection, system administration roles, security, backup strategies, and technical documentation. It also helped me develop better problem-solving and decision-making skills. As a BSIT student, I believe this experience will help me become a better System Administrator because it taught me to analyze an organization's needs, plan solutions carefully, document technical decisions, and consider both current requirements and future growth.

REFERENCES

Amazon Web Services. (n.d.). AWS Certified Cloud Practitioner. AWS Certified Cloud Practitioner
Canonical. (n.d.). Canonical Academy. Canonical Academy
Cisco. (n.d.). CCNA certification. CCNA certification
National Institute of Standards and Technology. (2020). Security and privacy controls for information systems and organizations (NIST Special Publication 800-53, Rev. 5). NIST SP 800-53 Rev. 5
Ubuntu. (n.d.). Ubuntu Server. Ubuntu Server